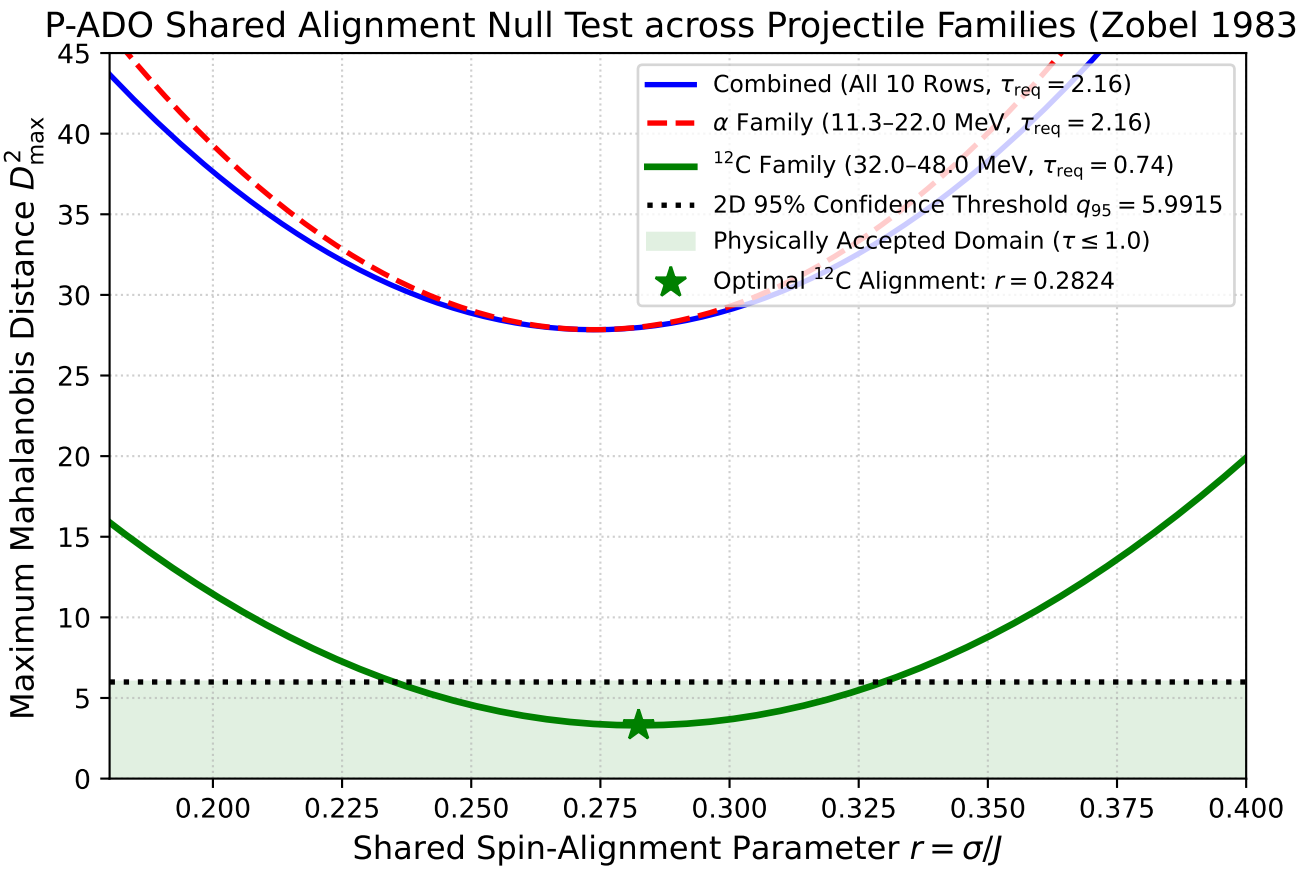
P-ADO Shared Alignment Null Test across Projectile Families (Zobel 1983)
Maximum Mahalanobis Distance D^2_{max}
Shared Spin-Alignment Parameter $r = \sigma/J$
Combined (All 10 Rows, $\tau_{req} = 2.16$)
α Family (11.3–22.0 MeV, $\tau_{req} = 2.16$)
^{12}C Family (32.0–48.0 MeV, $\tau_{req} = 0.74$)
2D 95% Confidence Threshold $q_{95} = 5.9915$
Physically Accepted Domain ($\tau \leq 1.0$)
Optimal ^{12}C Alignment: $r = 0.2824$
0
5
10
15
20
25
30
35
40
45
0.200
0.225
0.250
0.275
0.300
0.325
0.350
0.375
0.400